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Total Number of Pages : 02

**BTech
REC3C002**

3rd Semester Regular / Back Examination: 2021-22

SIGNALS & SYSTEMS

BRANCH(S): ETC / ECE

Time : 3 Hour

Max Marks : 100

Q.Code : OF802

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions : (2×10)

- a) Plot the signal $x(n) = 3u(n) - 2u(n-2) + 2$
- b) What do you mean by BIBO stable?
- c) What is normalized Autocorrelation sequence?
- d) Explain the associative property of linear convolution?
- e) State the duality property of Fourier transform?
- f) Write the trigonometric form of Fourier series of a periodic signal?
- g) Define ROC for one sided Z-transform of a signal?
- h) Find inverse Z-transform of the signal $X(z) = 5z^{-1} + 6z^{-4} - 8z^3 + 10$
- i) $x(n) = \{19, 8\}$, find its DFT ?
- j) What is DTFT and DFT, mention the relationship?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6×8)

- a) What is an energy signal, power signal, explain with examples?
- b) $y(n) = 3x(n) + 5$, is a LTI system or not, Justify ?
- c) Find the linear convolution between the sequences $\{2, 1, -3\}$ and $\{10, 8, 2\}$
- d) Find the cross correlation of the sequence $\{-2, 6, 7, 1, 9\}$ and $\{1, -2, 1\}$
- e) Write and prove the integration property of Fourier Series of a continuous time periodic signal?
- f) Find the Fourier transform of a triangular signal, with amplitude 5 v and duration 2T, symmetric about Y-axis.
- g) State and prove the correlation property of the Z-transform.
- h) $x(n) = \{2, 3, -4, 1, 5, 6, 2, 1\}$, find one sided Z-transform of $x(n+3)$?
- i) Find the Auto correlation of the sequence $\{1, 2, 5, 2, -1, 4\}$ using Z-transformation.
- j) Explain symmetry property of DFT.
- k) Find the DFT of the sequence $\{2+3j, 1, 2-2j, -3\}$
- l) Derive the condition of stability and causality in Z-domain.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3

Classify discrete time signals and systems in detail with examples.

(16)
- Q4

What is unit step response of a system? Find the impulse response of a LTI system described by the difference equation $y(n) = x(n) + 2x(n-3)$.

(16)
- Q5

Explain the various methods to find the inverse Z-transform of a signal, with merits and demerits?

(16)
- Q6

Given $x(n) = \{1, -2, 3, 2\}$, $y(n) = \{2, -1, 1, 2\}$, find its circular convolution using DFT, IDFT method ?

(16)